

Rafal Dlugopolski

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WORK EXPERIENCE

Fermilab (Fermi National Accelerator Laboratory)

Sep. 2023 - Present

Staff Engineer II

Batavia, IL

- RF Engineer – Part of the RF group consisting of scientists and engineers researching, designing, and building RF cavities. Currently designing and building a new 53Mhz recycler cavity for the new generation PIP-II accelerator upgrade.
- Assigned to multiple times to review engineering notes for sound engineering judgment and adherence to Fermilab and DOE standards.
- CAD Group Lead
 - As group lead, I lead 11 direct reports ranging from junior drafters to senior designers for the Accelerator Division in the machine support department. All work is done in Siemens NX and Teamcenter (PLM).
 - I interface with stakeholders in more complicated projects and when the need arises from direct reports facing sudden hurdles.

Sciaky, Inc.

Dec. 2019 – Sep. 2023

Mechanical Engineering Manager

Bedford Park, IL

- Sciaky, Inc. manufactures state of the art Electron Beam Welding and Additive (3D printing) machine tools.
- As Chief Mechanical Engineer and Manager, I oversee the mechanical engineering department (3-direct reports), lead all technical development, research, and manage all active programs.
 - Successfully completed various machine programs totaling \$35+ million in capital equipment delivered.
 - Report directly to Director of Operations and Engineering Manager.
- Interface with project teams and engineering groups to ensure seamless manufacturing and assembly.
- Design, create, and produce 3-D CAD parts, assembly and layout models, detail drawings, notes, assembly instructions and bill of material (BOM) using Creo 8.0 software.
- Collaborate with vendors to resolve manufacturing issues and conduct negotiations to ensure procurements are in line with overall project budget and schedule.

Master Machine & Repair

Feb. 2019 – Dec. 2019

Engineering Project Manager

Bedford Park, IL

- Managed a team of 12 mechanics, machinist, and welders in various projects run concurrently.
- Reverse engineered obsolete components to manufacture for emergency repairs.
- Designed and developed auxiliary systems for heavy industrial equipment (presses, mixers, conveyors) increasing production output.
- Produced 3-D CAD parts, assembly and layout models, detail drawings, notes, assembly instructions and bill of material (BOM) using SOLIDWORKS software.

EDM Zap Parts

May. 2016 – Feb. 2019

Mechanical Engineer

Lisle, IL

- Designed an EDM (electrical discharge machining) machine tool with 8 independent work heads to simultaneously cut 16 specimen coupons in 63Hrc Nickel superalloys for tensile testing.
- Responsible for development of motion systems on EDM sinker product line which included servo motors, linear rails, ball screws, and actuator sub-systems.
- Produced 3-D CAD parts, assembly and layout models, detail drawings, notes, assembly instructions and bill of material (BOM) using SOLIDWORKS software.

EDUCATION

Illinois Institute of Technology
Bachelor of Science in Mechanical Engineering

May, 2017
Chicago, IL

SKILLS & INTERESTS

- **Skills:** Project Management; Machine Design; High Vacuum Design; Prototyping; Design Reviews; Technical reports; DMFEA; DFM; ASME Y14.5 GD&T; FEA; SOLIDWORKS; PTC Creo; NX Siemens; Inventor; Ansys; Windchill; Teamcenter; PDM; BOM Management; ERP; Risk Management; MATLAB; Sensor Design; Microsoft Office.
- **Interests:** Puns; woodworking; machining; traveling; engineering; mechatronics; Reddit; the office; history; football.